

Frameworks

Frameworks

- The term “web framework” can mean many things, it depends upon who you ask.
- Agreement, a framework should make your job easier by providing code and structure.

What does “framework” mean here?

- Front-end developers
 - CSS, JavaScript, jQuery
- Back-end developers
 - Routing, resources, security

Popular front-end frameworks

- Bootstrap (2011)
 - Its popularity makes it... popular

What it means for you

- Many people build their sites directly from templates.
- Others use their own code and add functionality.
- Even “pure” coders should know the basics of one popular framework

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Introduction to Bootstrap

What is it?

- Framework for creating web sites
- Focuses on responsive, mobile-first approach
- Consists of
 - CSS and HTML templates
 - JavaScript extensions
- Emphasis on interfaces and layouts

Who uses it?

- Many popular sites are (or were) built using Bootstrap
 - [nba.com](https://www.nba.com)
 - [target.com](https://www.target.com)
 - [walmart.com](https://www.walmart.com)
 - [codeacademy.com](https://www.codecademy.com)
 - [bloomberg.com](https://www.bloomberg.com)
 - [airbnb](https://www.airbnb.com)

Bootstrap benefits

- 12-column grid system
 - Helps with spacing issues
 - Built-in responsive design
- Common jQuery functionalities
 - Accordion, Drop-down menus, Carousel
- Familiar “look and feel”
 - Many sites use Bootstrap
 - Makes your forms look “legitimate”

Why you should use it

- Fast development
- Platform Independent
- Responsive by default
- Customizable

Why you shouldn't use it....

- Doesn't follow best practices
 - Content and layout are intertwined
 - CSS frameworks and semantics
- Can be resource-heavy
- The look is somewhat generic
 - This can be good or bad.

What we will cover

- There are two ways to use Bootstrap
 - As a supplement to your style
 - As a theme that you expand upon
- We will talk about some of the basics, but there are still many parts left untouched.
- It is important to test often when using code that isn't yours.

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Getting Started With Bootstrap 3

Getting Started

- How do you actually use Bootstrap?
- You need access to the CSS code and JS code.
- Options:
 - Download copy
 - Use absolute reference
 - Link on CodePen

Saving your own copy of Bootstrap

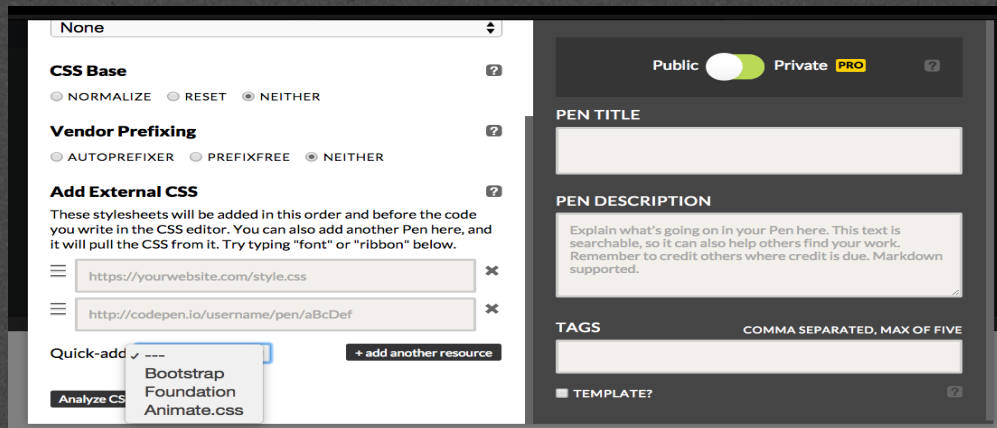
- You can download a copy of Bootstrap at <http://getbootstrap.com>
- Use the default values or select <http://getbootstrap.com/customize/>
- Make sure you know where you saved your files!

Using a CDN

- A CDN is a content delivery network
- Provides a way to connect to the Bootstrap code using an absolute reference
 - <http://getbootstrap.com/getting-started/>

Using Bootstrap with CodePen

- You can add Bootstrap to CodePen in the Settings.
- Select Settings → CSS → Add external CSS



Template

- It is common to start with a template file.

<http://getbootstrap.com/getting-started/#examples>

- A modified version from getbootstrap.com can be found here:

[RD:Bootstrap Template](#)

How do you know it is working?

- For such a simple file, it may be hard at first to see if it is working.
- To test:
 - Comment out link to css
 - Use Inspect Element
 - This method is definitely preferred, particularly as you start to add your own styles.

Don't forget...

- To use element inspector.
- To include the JavaScript at the bottom.
- That you can customize bootstrap defaults.
- That you can overwrite bootstrap defaults.



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Bootstrap Grid System

It's not a table....

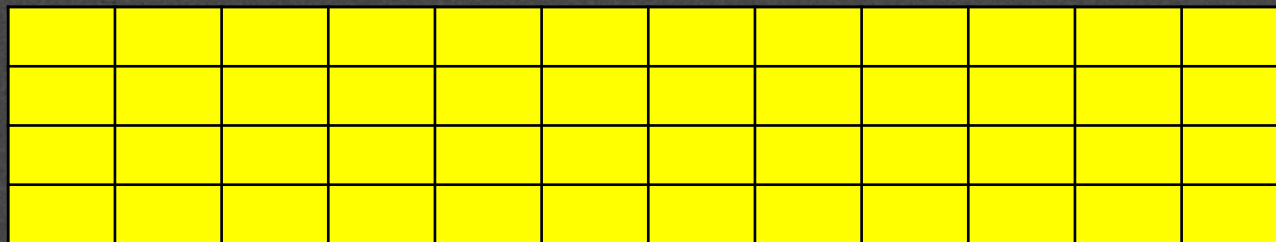
- The Bootstrap layout is based on a 12 column grid

12 columns = 100%

6 columns = 50%

9 columns = 75%

3 columns = 25%



Grid classes

- Every grid consists of:
 - A container
 - A row
 - One or more column classes

```
<div class= "container">
```

```
<div class= "row">
```

```
<div class= "col-xx-yy">
```


Setting width with col-xx-yy

- The class col-xx-yy always starts with “col”
- The “xx” component is the viewport size:
 - xs, sm, md, lg
- The “yy” component is the number of columns 0..12

col-xx-yy

- col-xs-12
- col-sm-8
- col-md-4
- col-lg-3
- These are often combined! Think photos....

```
<img src = "pic.jpg" class="col-xs-12 col-sm-6  
col-md-3 col-lg-2">
```


But it is actually easier than that...

- Because Bootstrap is mobile first, there is no need to specify the twelve columns if the element is block.

```
<div class="col-xs-12 col-md-3">Yellow Part</div>
```

```
<div class="col-md-3">Yellow Part</div>
```

Positioning classes

- On viewports md and up, there is an option for push and pull class.
 - **col-XX-push-YY** → move YY columns to the right
 - **col-XX-pull-YY** → move YY columns to the left

Responsive utility classes

- **hidden-XX** content will only be hidden on the XX screen size
- **visible-XX** content will only be visible on the XX screen size
- **sr-only** content is hidden on all devices except screen readers

Conclusion

- Bootstrap is mobile first, larger screens inherit values from smaller screens.
- Review <http://getbootstrap.com/examples/grid/>
- Do the math!! If you want two elements to be next to each other on a larger screen, make sure their width adds up to no more than 12.

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Bootstrap Navigation

Navigation Bars

- One of the components that gives Bootstrap its familiar “look and feel” is the navigation options.
- The nav class is combined with other classes to create each style

Making a navigation bar

- Decide what type of links you want:
 - nav-tabs vs nav-pills
 - [RD:nav-tabs](#)
 - [RD: nav-pills](#)
- Decide on layout (horizontal, stack, justified, etc.)
 - nav-stacked, nav-justified

Drop downs

- To add dropdown menus you need to include the Bootstrap js files AND a link to the jQuery.
- Bootstrap
example: <http://getbootstrap.com/components/#pills-with-dropdowns>

[RD:dropdowns](#)

navbar class

- The navbar class serves as a navigation header for your application or site.
- Positioning includes:
 - navbar-static-top
 - navbar-fixed-top
 - navbar-fixed-bottom

[RD:navbar](#)

Accessibility

- Using a “nav” class does not convey semantics.
- Use the `<nav>` tag or ARIA attribute `role=“navigation”`.

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Closing

What Have We Covered?

- Write basic CSS rules for styling.
- Utilize tools to check our CSS (Inspect Element, Wave, etc).
- Use fluid measurements.
- Write media queries for different screen sizes.

What Else?

- Write rules to facilitate accessibility with motion and color scheme preferences.
- Discuss the importance of the mobile-first paradigm.
- Understand the grid system of Bootstrap.

What Next?

- More practice until you feel more comfortable.
- Look more into frontend frameworks, eg React, Angular, etc.
- Explore accessibility in more depth.
- Use your knowledge to help elevate existing team projects.

Congratulations!

- You can also take a moment to be proud of yourself.

Great job!

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